

Precept 4

Goals

- Solve problems regarding the Bernoulli processes (with the use of Binomial and Geometric random variables).
- Review how to calculate the variance of a random variable, the variance of a sum in two cases: when summands are independent, and when summands are not independent.
- Use the idea of conditioning in order to compute expectations, variances, and probabilities.

Reminder

- **Expectation and Variance:** Let $X : \Omega \mapsto D$ be a discrete random variable.

Name	Parameters	Distribution	$\mathbb{E}[X]$	$\text{Var}(X)$
$X \sim \text{Bernoulli}$				
$X \sim \text{Binomial}$				
$X \sim \text{Geometric}$				
$X \sim \text{Poisson}$				

- **Law Of The Unconscious Statistician - Discrete version:** for a function φ , we have:

$$\mathbb{E}[\varphi(X)] = \sum_{x \in D} \varphi(x) \mathbb{P}(X = x) \quad (\text{LOTUS})$$

- **Variance and Covariance:** Let X, Y be random variables.

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - \mathbb{E}[X]^2$$

$$\forall a, b \in \mathbb{R}, \text{Var}(aX + b) = a^2 \text{Var}(X)$$

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$$

- **Variance of a sum:**

$$\text{Var}(X \pm Y) = \text{Var}(X) + \text{Var}(Y) \pm 2 \text{Cov}(X, Y)$$

If X and Y are independent, then $\text{Var}(X \pm Y) = \text{Var}(X) + \text{Var}(Y)$

- **Series Expansion of the Exponential :** $\forall x \in \mathbb{R}, \exp(x) = \sum_{k=0}^{\infty} \frac{x^k}{k!}$

Exercises

Exercise 1 (Sum of Bernoulli Variables - Binomial, Geometric, Hypergeometric distributions)

ORF309 has $n + m$ students, of which n are girls and m are boys. During a lecture, students are randomly selected, one at a time, and given an action figure.

- (a) Assume a student may be selected multiple times, i.e. we *sample with replacement* and each student is equally likely to be selected. K action figures are given out. Let X be the number of action figures given to girls. Find $\mathbb{E}[X]$.
- (b) Assume a student may be selected multiple times, i.e. we *sample with replacement* and each student is equally likely to be selected. Let Y be the number of action figures given to girls before the first boy is selected. Find $\mathbb{E}[Y]$. Assume that the number of figures is unlimited ($K = \infty$).
- (c) Assume that each student may be selected only once, i.e. we *sample without replacement* and each of the remaining students is equally likely to be selected. Let Z be the number of action figures given to girls before the first boy is selected. Find $\mathbb{E}[Z]$. Assume that $K \geq n$.

Exercise 2 (Variance of a Poisson random variable)

Let $\lambda > 0$ and $X \sim \text{Poisson}(\lambda)$. Prove that $\text{Var}(X) = \lambda$.

Exercise 3 (iPhone Headphone Jacks)

Three different factories in Asia are capable of making the new iPhone. The factories' names are A , B , and C and they are all of different quality. If factory A is given the newest iPhone contract they will accidentally add headphone jacks on a number of phones. For the duration of one week, this number has a Poisson distribution with mean $a = 100$. If factory B does the job the parameter is $b = 200$ and if factory C does the job the parameter is $c = 300$. Suppose that each factory is equally likely to be chosen.

- (a) Let X be the number of iPhones that receive headphone jacks throughout the year. Suppose each week Apple places an order independently across the three factories. Compute the mean and variance of X .
- (b) By the end of the first week no iPhones are found to be defective. What is the likelihood that Factory A received the contract?

Exercise 4 (Returning exams - time permitting)

Students of a class whose concentration and course code must remain anonymous have written their midterm exam and are anxiously awaiting for their results. Finally, an e-mail arrives telling them to pick up their exams from their individual folders. Unfortunately, the AI who was supposed to distribute their exams was running on close to zero sleep and has ended up placing the exams in a completely random order. If the class has 140 students, how many of them on average are expected to get their exams, and what is the variance of this number?